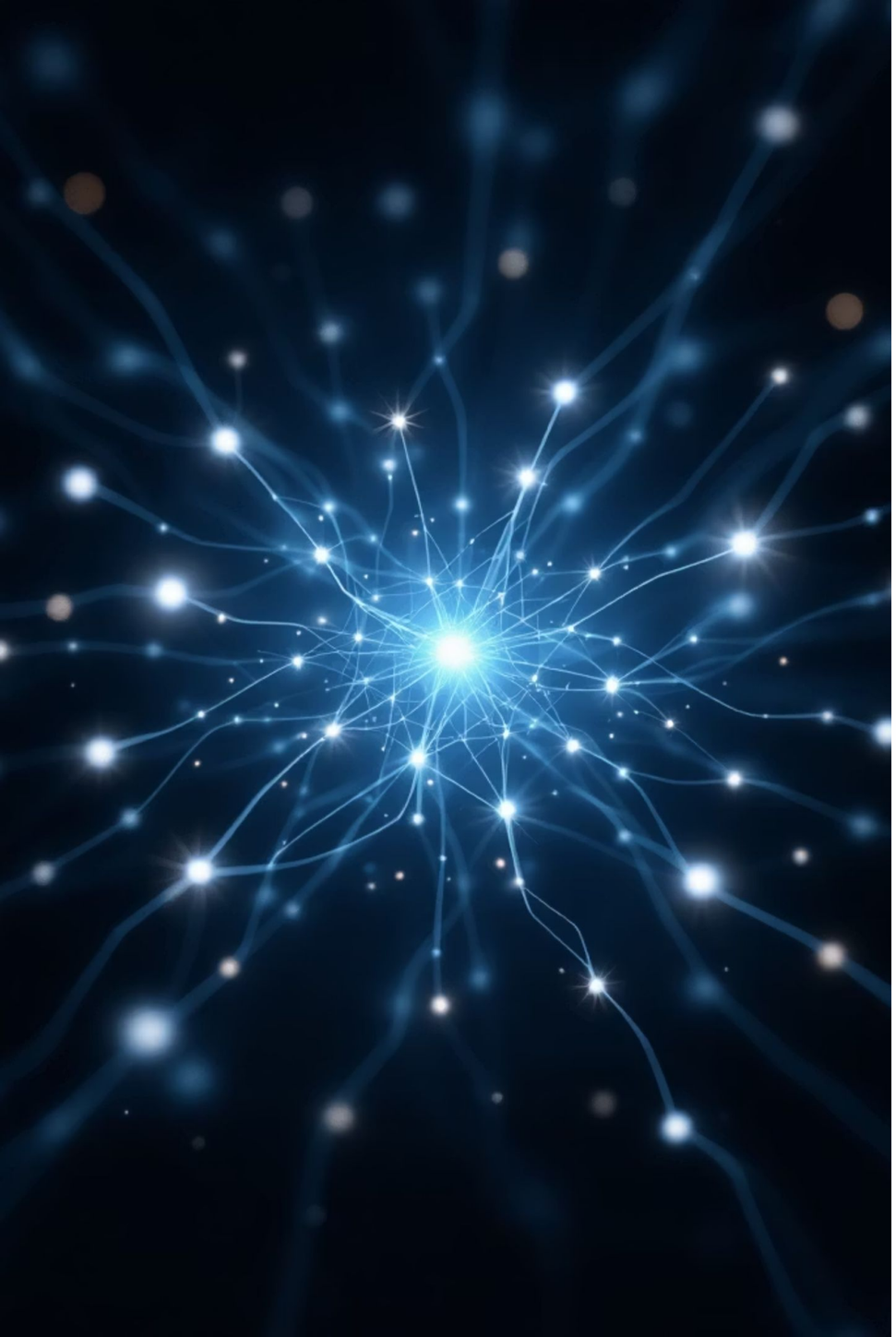
AI Cost Monitoring

MODULE 3.3.3

A practical look at how AI language models are accessed and used, covering API integration, model pricing structures, and how dashboards are used to monitor and control usage costs.



Using Large Language Models

Large language models (LLMs) are powerful tools that accept natural language prompts and return intelligent answers for example, asking **"What is a CVE?"** and receiving a detailed explanation instantly.

Cybersecurity Context

Useful for generating security content, asking questions about log entries, and analyzing vulnerabilities

Data Scanning

Scan and interpret large volumes of data quickly, surfacing relevant insights from complex inputs

Basic Interface

Users type prompts and receive responses directly on screen — simple, interactive, and immediate

Accessing Models via API

The real power of LLMs comes when accessed programmatically via an API enabling seamless integration with existing tools and workflows.

Without API

Manually copy and paste log events into the model interface, then manually interpret and act on the response — slow, error-prone, and not scalable.

→ Copy log event manually

→ Paste into model UI

→ Manually act on response

With API

Log events are sent directly to the model via the API. The enriched response can automatically trigger next steps generating alerts or creating firewall rules without human intervention.

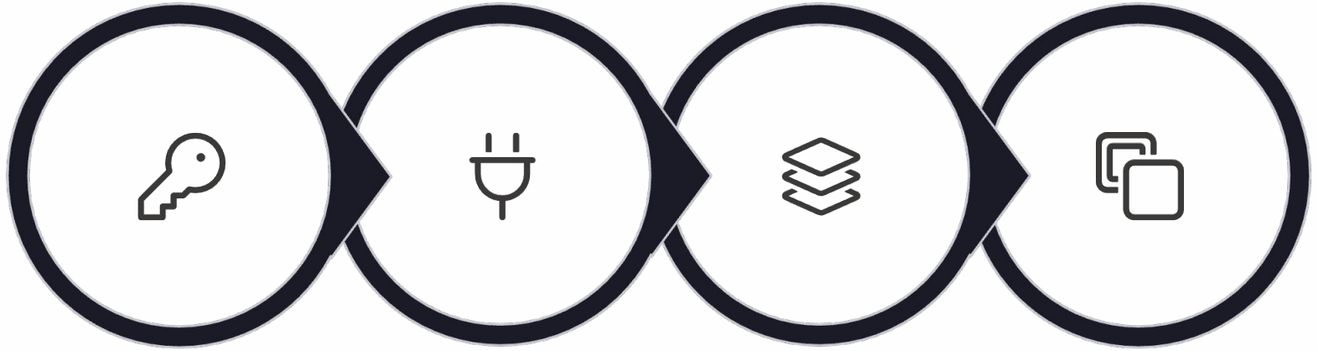
→ Log sent directly via API

→ Model enriches log data

→ Automated alerts & rules

How API Integration Works

An **API key** is what enables programmatic access to a large language model. It acts as a credential that links your tool to the provider's hosted model and to your billing account.



Get API Key | Plug Into Tool | Select Model | Send Request

Tools like Open Web UI connect to hosted models (e.g., GPT models hosted by OpenAI) using this key. When a user selects a model and submits a prompt, the tool sends the request to the LLM via the API and displays the response all transparently and automatically.

Model Pricing and Controlling Access

Different models carry very different price tags. More sophisticated, larger, or newer models tend to cost significantly more and cost is tied to both the **volume of information sent** and the **complexity of outputs returned**.

Premium Model

~$5.00 per use case — advanced reasoning, larger context windows, richer outputs

Economy Model

~$0.10 per use case — suitable for simpler tasks, faster responses, lower overhead

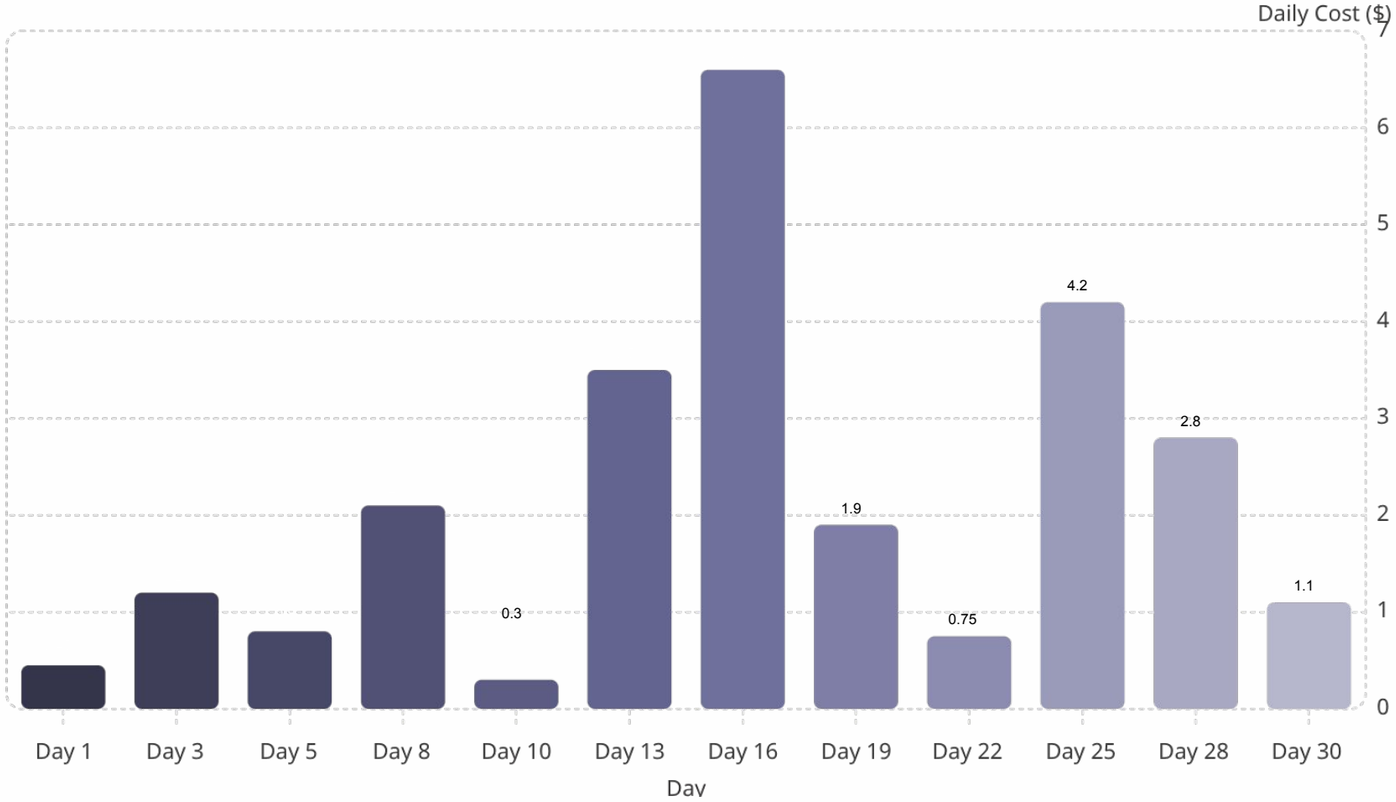
Restricting API Key Access

When setting up an API key, administrators can restrict which models that key is allowed to access.

- Prevents employees from using expensive models unnecessarily
- Keeps costs aligned with the task at hand
- Enforces governance over AI tool usage
- Reduces risk of unexpected billing spikes

Monitoring Usage with a Dashboard

API providers like OpenAI offer dashboards that show exactly how an API key has been used and the cost associated with that usage over time, broken down day by day.



Usage can vary significantly across a 30-day span — some days may be under a dollar, while peak days may reach **$6.60 or more**. This visibility helps organizations understand whether usage aligns with expectations and budget.

Making Cost-Based Decisions

Dashboard insights transform raw usage data into actionable decisions. Organizations can use this visibility to optimize both cost and capability.

1

Set Spending Limits

Ask: Do we need a daily, weekly, or monthly cap? Use dashboard trends to set thresholds that prevent budget overruns before they happen.

2

Switch to a Cheaper Model

If the chosen model is costing more than anticipated, consider switching to a more economical alternative that still meets the task requirements.

3

Upgrade When Budget Allows

If usage is well under budget, a more sophisticated model could be adopted, unlocking better outputs without exceeding financial limits.

The dashboard is the key tool for informed decision-making around AI cost management treat it as an essential operational resource, not an afterthought.

Key Takeaways

Interactive & Programmatic

LLMs can be used via prompts directly or integrated programmatically via an API for powerful automation

API Enables Automation

Logs, alerts, and actions can be connected directly to AI responses, eliminating manual steps

Manage API Keys Carefully

API keys enable access and are tied to billing — restrict model access to control spending and governance

Model Costs Vary Widely

Different models carry different costs — restrict API key access to appropriate models to avoid unnecessary spend

Day-by-Day Visibility

Usage dashboards provide granular cost visibility, enabling budget enforcement and smarter model selection

Ongoing Process

Monitoring AI costs is continuous — dashboards are essential for staying within budget as usage evolves